Clock sinks from placement

Every CTS algorithm starts from sinks, cell centers that must receive the clock, and one source

The idea

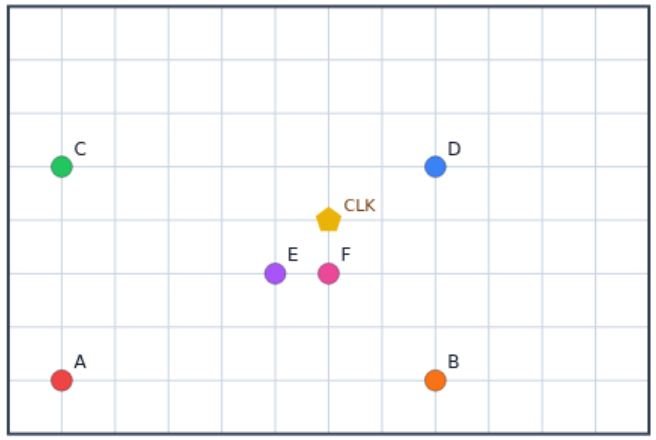
- Sink_points reads placement for A through F
- Source sits at six comma four
- Moving a sink moves its access point for the next tree build

12×8 chip

CLOCK SINKS

12×8 chip

Step 1 / 5 · chip · module01-01-clock-sinks



Explanation

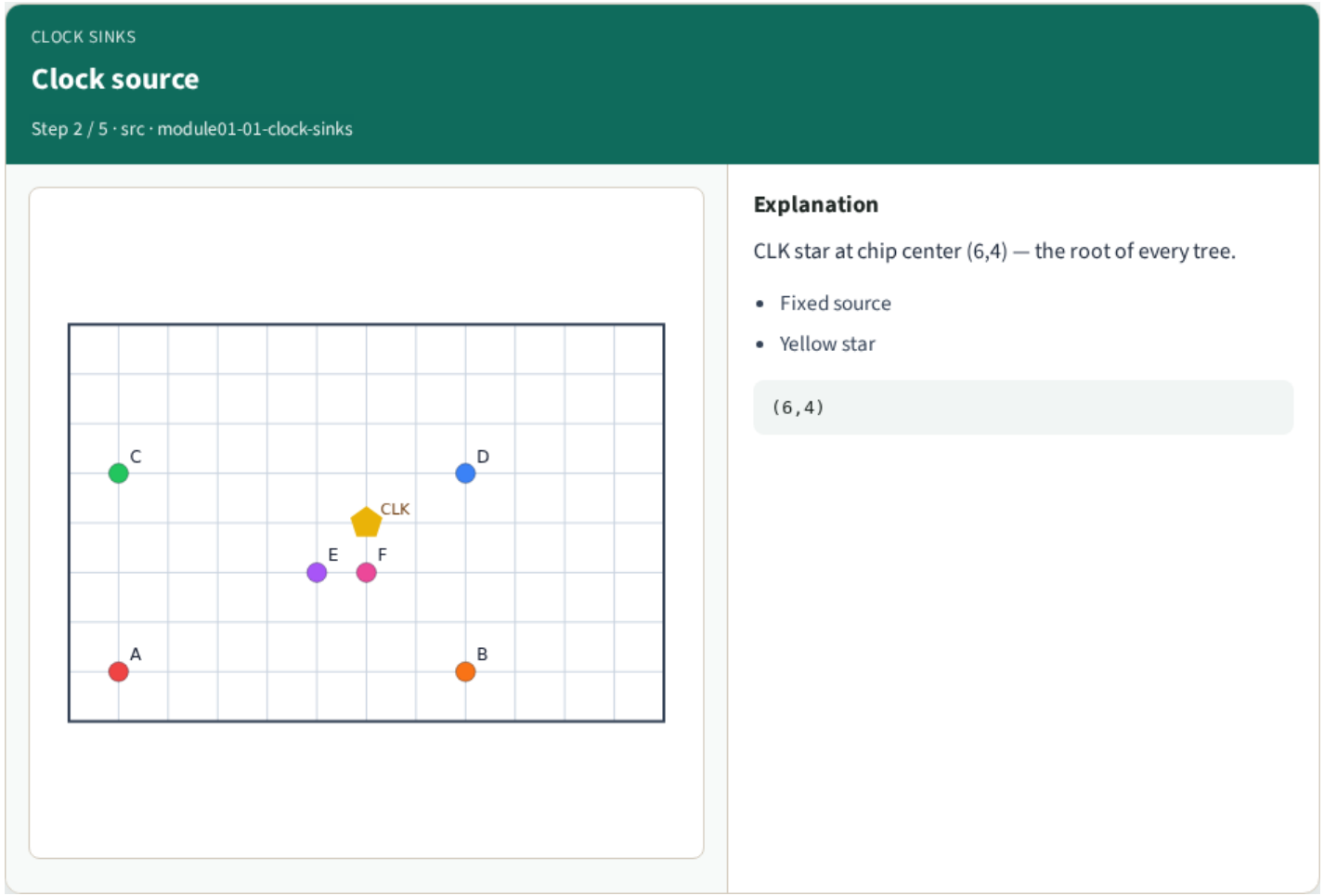
Six sinks A–F sit on the same spread placement as routing labs.

- Same PD spine
- Toy CTS

```
6 sinks
```

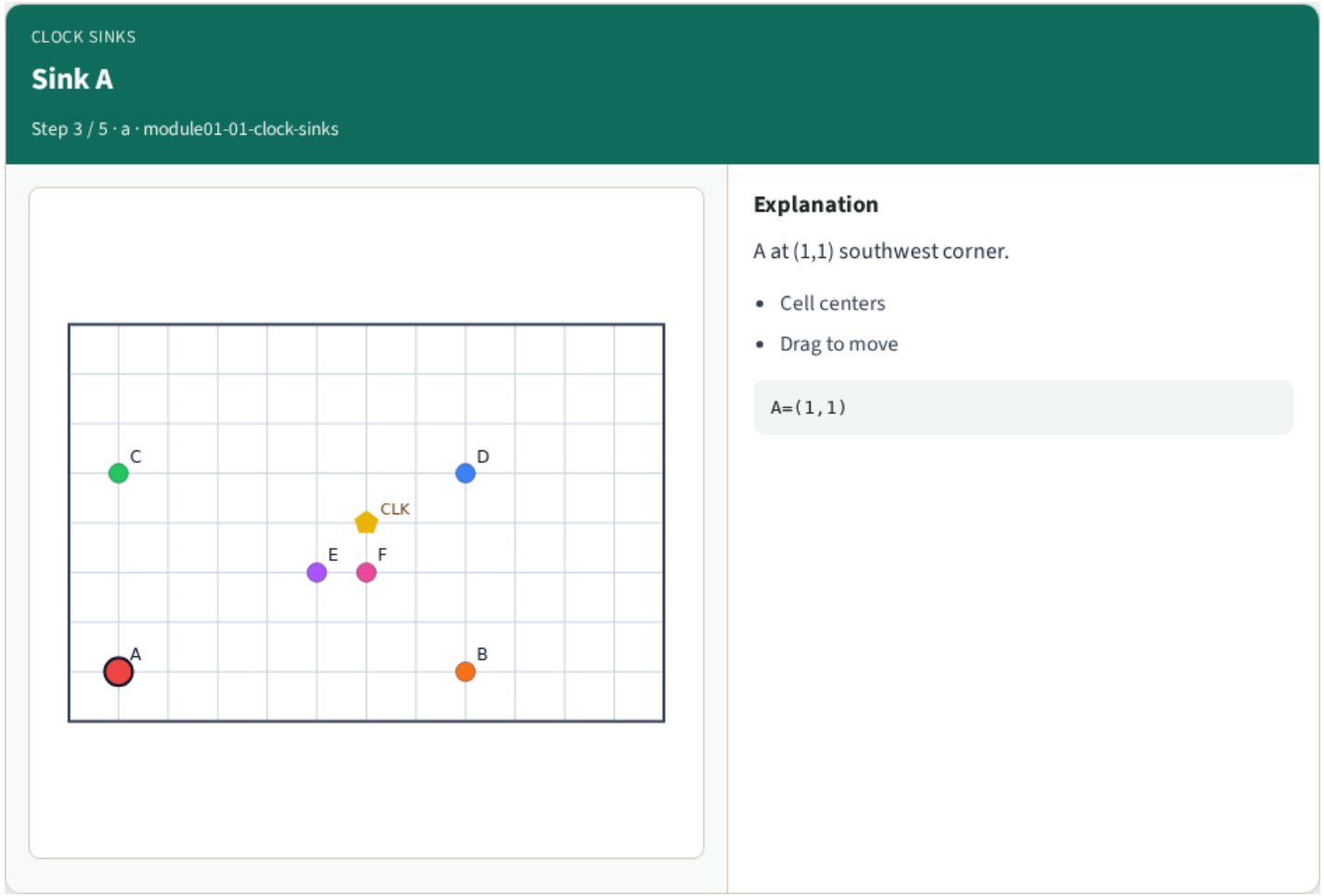
12×8 chip

Clock source



Clock source

Sink A


CLOCK SINKS
Sink A
Step 3 / 5 · a · module01-01-clock-sinks
C
D
CLK
E
F
A
B
Explanation
A at (1,1) southwest corner.
Cell centers
Drag to move
A=(1,1)

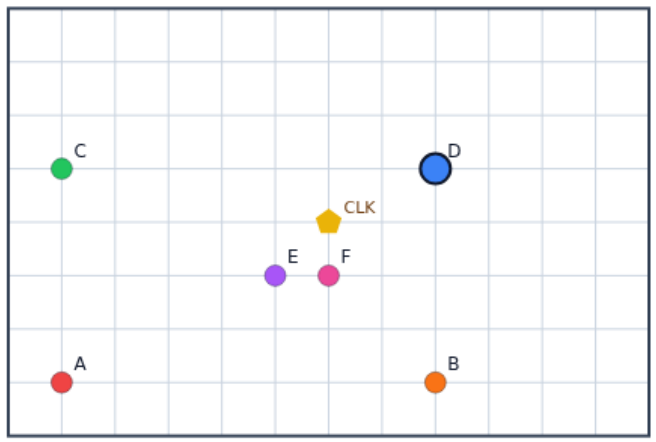
Sink A

Sink D

CLOCK SINKS

Sink D

Step 4 / 5 · d · module01-01-clock-sinks



Explanation

D at (8,5) northeast.

- Opposite corner
- Long path to A

```
D=(8,5)
```

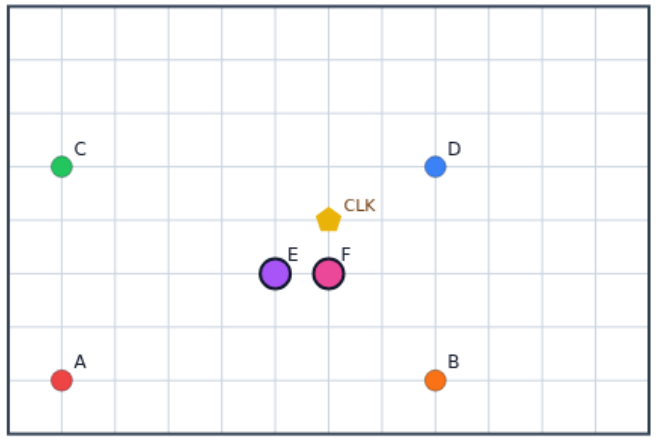
Sink D

All six

CLOCK SINKS

All six

Step 5 / 5 · all · module01-01-clock-sinks



Explanation

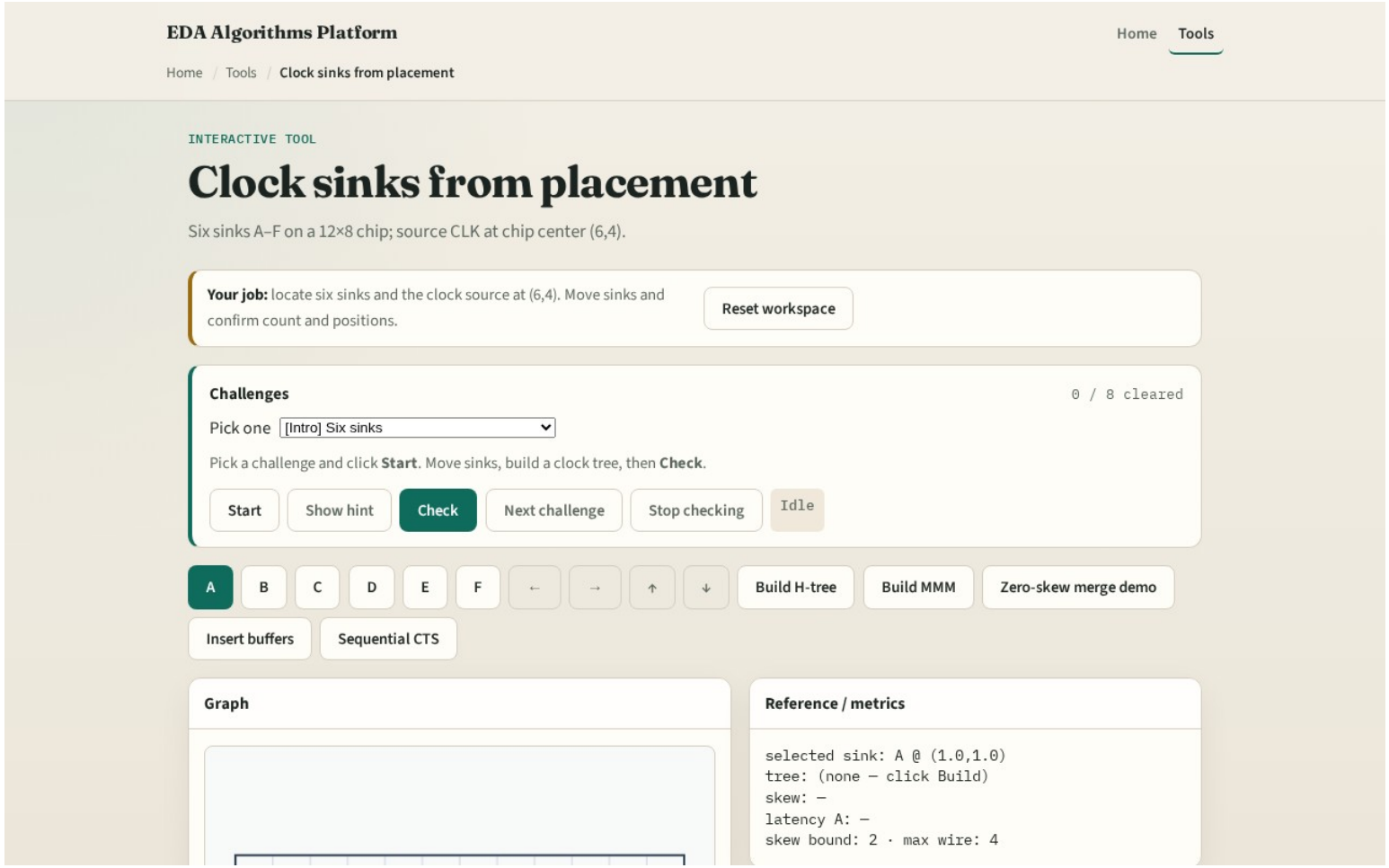
E/F sit near the center corridor beside the source.

- E≈(5,3)
- F≈(6,3)

```
Ready for trees
```

All six

Browser lab track



Browser lab starter

Implement track

- Call `sink_points(placement, sinks)` in `common/ctsutil.py`
- Print A through F on tiny_cts
- Assert source at (6,4)

Pitfalls

- Using GCell indices instead of cell centers
- Forgetting the source is fixed
- Treating nets as sinks

Your turn

- Next: measure latency and skew on a built tree